Singular Homology

A **singular n-simplex** in a space X is by definition just a map $\sigma : \Delta^n \to X$. The word 'singular' is used here to express the idea that σ need not be a nice embedding but can have 'singularities' where its image does not look at all like a simplex. All that is required is that σ be continuous. Let $C_n(X)$ be the free abelian group with basis the set of singular n-simplices in X. Elements of $C_n(X)$, called **n-chains**, or more precisely singular n-chains, are finite formal sums $\sum_i n_i\sigma_i$ for $n_i \in \mathbb{Z}$ and $\sigma_i : \Delta^n \to X$. A boundary map $\partial_n : C_n(X) \to C_{n-1}(X)$ is defined by the same formula as before:

$$\partial_n(\sigma) = \sum_i (-1)^i \sigma \,|\, [v_0, \cdots, \hat{v}_i, \cdots, v_n]$$

Implicit in this formula is the canonical identification of $[v_0, \cdots, \hat{v}_i, \cdots, v_n]$ with Δ^{n-1}, preserving the ordering of vertices, so that $\sigma \,|\, [v_0, \cdots, \hat{v}_i, \cdots, v_n]$ is regarded as a map $\Delta^{n-1} \to X$, that is, a singular $(n-1)$-simplex.

Often we write the boundary map ∂_n from $C_n(X)$ to $C_{n-1}(X)$ simply as ∂ when this does not lead to serious ambiguities. The proof of Lemma 2.1 applies equally well to singular simplices, showing that $\partial_n\partial_{n+1} = 0$ or more concisely $\partial^2 = 0$, so we can define the **singular homology group** $H_n(X) = \operatorname{Ker}\partial_n / \operatorname{Im}\partial_{n+1}$.

It is evident from the definition that homeomorphic spaces have isomorphic singular homology groups H_n, in contrast with the situation for H_n^Δ. On the other hand, since the groups $C_n(X)$ are so large, the number of singular n-simplices in X usually being uncountable, it is not at all clear that for a Δ-complex X with finitely many simplices, $H_n(X)$ should be finitely generated for all n, or that $H_n(X)$ should be zero for n larger than the dimension of X — two properties that are trivial for $H_n^\Delta(X)$.

Though singular homology looks so much more general than simplicial homology, it can actually be regarded as a special case of simplicial homology by means of the following construction. For an arbitrary space X, define the **singular complex** $S(X)$ to be the Δ-complex with one n-simplex Δ_σ^n for each singular n-simplex $\sigma : \Delta^n \to X$, with Δ_σ^n attached in the obvious way to the $(n-1)$-simplices of $S(X)$ that are the restrictions of σ to the various $(n-1)$-simplices in $\partial\Delta^n$. It is clear from the definitions that $H_n^\Delta(S(X))$ is identical with $H_n(X)$ for all n, and in this sense the singular homology group $H_n(X)$ is a special case of a simplicial homology group. One can regard $S(X)$ as a Δ-complex model for X, although it is usually an extremely large object compared to X.

Cycles in singular homology are defined algebraically, but they can be given a somewhat more geometric interpretation in terms of maps from finite Δ-complexes. To see this, note first that a singular n-chain ξ can always be written in the form $\sum_i \varepsilon_i\sigma_i$ with $\varepsilon_i = \pm 1$, allowing repetitions of the singular n-simplices σ_i. Given such an n-chain $\xi = \sum_i \varepsilon_i\sigma_i$, when we compute $\partial\xi$ as a sum of singular $(n-1)$-simplices with signs ± 1, there may be some *canceling pairs* consisting of two identical singular $(n-1)$-simplices with opposite signs. Choosing a maximal collection of such canceling pairs, construct an n-dimensional Δ-complex K_ξ from a disjoint union of n-simplices Δ_i^n, one for each σ_i, by identifying the pairs of $(n-1)$-dimensional faces corresponding to the chosen canceling pairs. The σ_i's then induce a map $K_\xi \to X$. If ξ is a cycle, all the $(n-1)$-simplices of K_ξ come from canceling pairs, hence are faces of exactly two n-simplices of K_ξ. Thus K_ξ is a manifold, locally homeomorphic to $\mathbb{R}^n$, except at a subcomplex of dimension at most $n-2$. All the n-simplices of K_ξ can be coherently oriented by taking the signs of the σ_i's into account, so K_ξ is actually an oriented manifold away from its nonmanifold points. A closer inspection shows that K_ξ is also a manifold near points in the interiors of $(n-2)$-simplices, so the nonmanifold points of K_ξ in fact have dimension at most $n-3$. However, near the interiors of $(n-3)$-simplices it can very well happen that K_ξ is not a manifold.

In particular, elements of $H_1(X)$ are represented by collections of oriented loops in X, and elements of $H_2(X)$ are represented by maps of closed oriented surfaces into X. With a bit more work it can be shown that an oriented 1-cycle $\coprod_\alpha S^1_\alpha \to X$ is zero in $H_1(X)$ iff it extends to a map of an oriented surface into X, and there is an analogous statement for 2-cycles. In the early days of homology theory it may have been believed, or at least hoped, that this close connection with manifolds continued in all higher dimensions, but this has turned out not to be the case. There is a sort of homology theory built from manifolds, called *bordism*, but it is quite a bit more complicated than the homology theory we are studying here.

After these preliminary remarks let us begin to see what can be proved about singular homology.

Proposition 2.6. *Corresponding to the decomposition of a space X into its path-components X_α there is an isomorphism of $H_n(X)$ with the direct sum $\bigoplus_\alpha H_n(X_\alpha)$.*

Proof: Since a singular simplex always has path-connected image, $C_n(X)$ splits as the direct sum of its subgroups $C_n(X_\alpha)$. The boundary maps ∂_n preserve this direct sum decomposition, taking $C_n(X_\alpha)$ to $C_{n-1}(X_\alpha)$, so $\operatorname{Ker}\partial_n$ and $\operatorname{Im}\partial_{n+1}$ split similarly as direct sums, hence the homology groups also split, $H_n(X) \approx \bigoplus_\alpha H_n(X_\alpha)$. ◻

Proposition 2.7. *If X is nonempty and path-connected, then $H_0(X) \approx \mathbb{Z}$. Hence for any space X, $H_0(X)$ is a direct sum of $\mathbb{Z}$'s, one for each path-component of X.*

Proof: By definition, $H_0(X) = C_0(X)/\operatorname{Im}\partial_1$ since $\partial_0 = 0$. Define a homomorphism $\varepsilon: C_0(X) \to \mathbb{Z}$ by $\varepsilon(\sum_i n_i\sigma_i) = \sum_i n_i$. This is obviously surjective if X is nonempty. The claim is that $\operatorname{Ker}\varepsilon = \operatorname{Im}\partial_1$ if X is path-connected, and hence ε induces an isomorphism $H_0(X) \approx \mathbb{Z}$.

To verify the claim, observe first that $\operatorname{Im}\partial_1 \subset \operatorname{Ker}\varepsilon$ since for a singular 1-simplex $\sigma: \Delta^1 \to X$ we have $\varepsilon\partial_1(\sigma) = \varepsilon(\sigma|[v_1] - \sigma|[v_0]) = 1 - 1 = 0$. For the reverse inclusion $\operatorname{Ker}\varepsilon \subset \operatorname{Im}\partial_1$, suppose $\varepsilon(\sum_i n_i\sigma_i) = 0$, so $\sum_i n_i = 0$. The σ_i's are singular 0-simplices, which are simply points of X. Choose a path $\tau_i: I \to X$ from a basepoint

x_0 to $\sigma_i(v_0)$ and let σ_0 be the singular 0-simplex with image x_0. We can view τ_i as a singular 1-simplex, a map $\tau_i:[v_0,v_1]\to X$, and then we have $\partial\tau_i = \sigma_i - \sigma_0$. Hence $\partial(\sum_i n_i\tau_i) = \sum_i n_i\sigma_i - \sum_i n_i\sigma_0 = \sum_i n_i\sigma_i$ since $\sum_i n_i = 0$. Thus $\sum_i n_i\sigma_i$ is a boundary, which shows that $\operatorname{Ker}\varepsilon \subset \operatorname{Im}\partial_1$. $\square$

Proposition 2.8. *If X is a point, then $H_n(X) = 0$ for $n > 0$ and $H_0(X) \approx \mathbb{Z}$.*

Proof: In this case there is a unique singular n-simplex σ_n for each n, and $\partial(\sigma_n) = \sum_i(-1)^i\sigma_{n-1}$, a sum of $n+1$ terms, which is therefore 0 for n odd and σ_{n-1} for n even, $n \neq 0$. Thus we have the chain complex

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\approx} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\approx} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

with boundary maps alternately isomorphisms and trivial maps, except at the last $\mathbb{Z}$. The homology groups of this complex are trivial except for $H_0 \approx \mathbb{Z}$. $\square$

It is often very convenient to have a slightly modified version of homology for which a point has trivial homology groups in all dimensions, including zero. This is done by defining the **reduced homology groups** $\widetilde{H}_n(X)$ to be the homology groups of the augmented chain complex

$$\cdots \longrightarrow C_2(X) \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

where $\varepsilon(\sum_i n_i\sigma_i) = \sum_i n_i$ as in the proof of Proposition 2.7. Here we had better require X to be nonempty, to avoid having a nontrivial homology group in dimension -1. Since $\varepsilon\partial_1 = 0$, ε vanishes on $\operatorname{Im}\partial_1$ and hence induces a map $H_0(X)\to\mathbb{Z}$ with kernel $\widetilde{H}_0(X)$, so $H_0(X) \approx \widetilde{H}_0(X)\oplus\mathbb{Z}$. Obviously $H_n(X) \approx \widetilde{H}_n(X)$ for $n > 0$.

Formally, one can think of the extra $\mathbb{Z}$ in the augmented chain complex as generated by the unique map $[\varnothing]\to X$ where $[\varnothing]$ is the empty simplex, with no vertices. The augmentation map ε is then the usual boundary map since $\partial[v_0] = [\hat{v}_0] = [\varnothing]$.

Readers who know about the fundamental group $\pi_1(X)$ may wish to make a detour here to look at §2.A where it is shown that $H_1(X)$ is the abelianization of $\pi_1(X)$ whenever X is path-connected. This result will not be needed elsewhere in the chapter, however.

Homotopy Invariance

The first substantial result we will prove about singular homology is that homotopy equivalent spaces have isomorphic homology groups. This will be done by showing that a map $f:X\to Y$ induces a homomorphism $f_*:H_n(X)\to H_n(Y)$ for each n, and that f_* is an isomorphism if f is a homotopy equivalence.

For a map $f:X\to Y$, an induced homomorphism $f_\sharp:C_n(X)\to C_n(Y)$ is defined by composing each singular n-simplex $\sigma:\Delta^n\to X$ with f to get a singular n-simplex